

Final Project task

Social Network Analysis 2026, deadline: 4 March

The purpose of this final project is to go through typical steps of longitudinal social network analysis from data exploration through analysis till drawing conclusions from your results. You are requested to use SAOMs for the final project with **at least three waves of data and at least one chosen classroom**. You have a limited freedom to choose your own research question and the particular classroom or classrooms. The data is collected by Andrea Knecht and was part of her dissertation:

Knecht, A. (2006), *Networks and actor attributes in early adolescence [2003/04]*, Utrecht, The Netherlands Research School ICS, Department of Sociology, Utrecht University (ICS-Codebook no. 61).

The codebook for the data is available from the Siena webpage:

<https://www.stats.ox.ac.uk/~snijders/siena/>

The full dataset is available from the DANS archive after registration and requesting access:

<https://ssh.datastations.nl/dataset.xhtml?persistentId=doi:10.17026/DANS-Z9B-H2BP>

Cite this data source as:

Knecht, A. (2004). Network and actor attributes in early adolescence, <https://doi.org/10.17026/DANS-Z9B-H2BP>, DANS Data Station Social Sciences and Humanities.

You can find the student-reported network data in the Final Project Task folder. The data contains full classroom networks in sav (SPSS) format, which you can import, for instance, by `read_sav()`, which is a function in package `haven`. There are 126 school classes from 14 schools in 4 waves. The waves are labeled as V, W, X, and Y. There are multiple network layers (practical support, emotional support, leisure time contact, etc.). These are explained in the codebook. In each layer, a maximum of 12 classmates could have been nominated. Responses in the sav file are coded one by one into different variables, for instance, the responses to the network variable with label *Best friends of respondent* are included in variables *friend1*, *friend2*, *friend3*, ..., *friend12*.

There are multiple student attributes you might use. You can find more information in the codebook.

1. Consider “Best friends” as your dependent network. First, transform responses into a format that you will use for social network analysis. Describe the basic characteristics of this network over multiple waves for the selected classrooms. Please include formatted summary tables in the main text. Check the amount of change between waves and the stability of the composition of node set. Based on these descriptives, choose this or - if needed because of the descriptives - another classroom that you explore first for further investigation. Explain your choice with arguments.

Hint: In the literature seminar readings concerning SAOMs, you find examples why a classroom or an organization unit is dropped from an analysis (e.g., look at composition change, the Jaccard index, insufficient response).

2. Formulate one hypothesis of a dyadic character. Please justify the hypothesis with arguments from the literature or based on theoretical considerations.

3. Formulate one hypothesis of a triadic character (transitivity or cyclicity). Please justify the hypothesis with arguments from the literature or based on theoretical considerations.

4. Consider at least one node-level characteristic and formulate a hypothesis that links this characteristic and the formation and maintenance of best friendship nominations.

5. List the variables (ego, alter, dyadic, structural) you control for and include at least one sentence of explanation for each variable you include in the analysis.

You are asked to test your hypotheses in one classroom. You are advised to use three waves, but you might use all four waves. If you choose less than four waves, explain the choice of selection.

6. Visualize your dependent network over time in the selected classroom with using node colors for the selected node-level characteristic and FIXED spatial positions of the nodes.

7. Operationalize your hypotheses for SAOM.

8. Conduct RSiena analyses for your hypotheses. If convergence is not reached, come up with a modified model and explain your choice. After multiple failed attempts, you are allowed to pick another classroom. Continue your analysis until convergence is reached. You **need to have** a

converging model at the end.

9. Check the goodness of fit (GoF). Explain what the GoF shows. Update your model at least once to obtain better fit if necessary.

10. Interpret your results. Discuss your findings and draw conclusions about your hypotheses. Summarize the limitations of your approach and speculate about how further improvements could be made.

It is requested that you hand in:

1, a pdf document that contains your answers, formatted figures and tables,

2, an annotated R script / Rmd / Qmd file,

(Please do not forget to comment your code – this is always a good practice)

3, data file for the chosen classroom in case you made any modifications (this is not necessary if your R script runs properly on the original data).

Please send your submission to karoly.takacs@liu.se no later than 4 March, Wednesday, 23:59 CET.